Hyperspectral Endmember Extraction by $(\mu + \lambda)$ Multiobjective Differential Evolution Algorithm Based on Ranking Multiple Mutations

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***Abstract*— Endmember extraction (EE) plays a crucial part in the hyperspectral unmixing (HU) process. To obtain satisfactory EE results, the EE can be considered as the multiobjective optimization problem to optimize the volume maximization (VM) and root-mean-square error (RMSE) simultaneously. However, it is often quite challenging to balance the conflict of these objectives. In order to tackle the challenges of multiobjective EE, we present a $(\mu + \lambda)$ multiobjective differential evolution algorithm ($(\mu + \lambda)$-MODE) based on ranking multiple mutations. In the $(\mu + \lambda)$-MODE algorithm, ranking multiple mutations are adopted to create the mutant vectors via the scaling factor pool to enhance the population diversity. Moreover, mutant vectors employ the binary crossover operator to generate the trial vectors through a crossover control parameter pool in $(\mu + \lambda)$-MODE to take advantage of the good information of the population. In addition, $(\mu + \lambda)$-MODE utilizes the fast nondominated sorting approach to sort the parent and trial vectors, and then selects the elitism offspring as the next population via the $(\mu + \lambda)$ selection strategy. Eventually, experimental comparative results in three real HSIs reveal that our proposed $(\mu + \lambda)$-MODE is superior to other EE methods.**

***Index Terms*— Endmember extraction (EE), multiobjective differential evolution (DE), ranking multiple mutations, $(\mu + \lambda)$ selection strategy.**

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I. Introduction

FOR the research of earth observation, hyperspectral images (HSIs) are extremely significant [1]–[3]. Unfortunately, the HSIs have issues of mixed pixels that can cause an inaccurate analysis result of the HSIs. Hyperspectral unmixing (HU) is a valid technique for solving mixed pixel problems, which aims to identify the pure spectrum set (endmembers) and the corresponding fractions (abundances) [4]–[7].

Endmember extraction (EE) plays a vital part in the HU process [8], [9]. The linear spectral mixture model (LSMM) becomes the popular spectral mixture model in the EE phase [10]. Recently numerous traditional LSMM-based EE algorithms have been proposed by many researchers such as N-FINDR [11], the pixel purity index (PPI) [12], and vertex component analysis (VCA) [13]–[15]. N-FINDR is introduced by Winter [11] to select the simplex of the largest volume for autonomous spectral EE. The PPI algorithm utilizes the minimum-noise fraction (MNF) to reduce dimensionality and then extracts endmember by projecting spectral vectors onto random vectors [12]. The novel endmembers in VCA are determined by the projection extreme until the endmember number is reached [13]. Three common conventional EE approaches are VCA, PPI, and N-FINDR, but these traditional EE algorithms are often sensitive to outlier and noise [16].

Nowadays researchers pay more attention to the evolutionary computation (EC) field for the EE problem to overcome the aforementioned disadvantages. A number of evolutionary algorithms (EAs) have been introduced recently in order to consider the EE problem as a combinational optimization problem. To achieve the lowest root-mean-square error (RMSE), Zhang *et al.* [17] implemented a discrete particle swarm optimization (DPSO) algorithm that extended PSO into discrete space. And Zhang *et al.* [18] also implemented an ant colony optimization (ACO) algorithm for EE handling. Inspired by quantum behavior, Liu *et al.* [19] introduced a novel quantum-behaved PSO (QPSO) algorithm with a cooperative approach to search endmembers via optimizing objectives of the simplex volume maximum and the RMSE minimum, respectively. A novel mutation operator accelerated QPSO (called MOAQPSO), which adopted a mutation operation to enhance the diversity for escaping the local optimum, was

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proposed by Xu *et al.* [20] to find endmembers. Owing to the efficient differential evolution (DE) algorithm for continuous optimization [21]–[23], Zhong *et al.* [24] suggested an adaptive DE (ADEE) approach for EE to achieve the RMSE minimization in the feasible domain. Many researchers have recently viewed the EE as the multiobjective optimization problem (MOP) and proposed multiobjective EAs (MOEAs) to address EE. Liu *et al.* [25] provided a multiobjective DPSO (MODPSO) for optimizing two objectives called the RMSE minimization and the volume maximization (VM) simultaneously. Through the archive strategy and the restart mechanism, Tong *et al.* [26] further proposed an improved MODPSO (IMODPSO) to enhance the Pareto quality of EE result.

From our work, a $(\mu + \lambda)$ multiobjective DE method based on ranking multiple mutations ($(\mu + \lambda)$-MODE) is proposed for hyperspectral EE. The EE problem is considered as the MOP in which the RMSE and the volume inverse can be minimized simultaneously. In our proposed $(\mu + \lambda)$-MODE, ranking multiple mutations are utilized to generate the mutant vectors through a scaling pool. Ranking multiple mutations mean that some parent vectors are proportionally selected based on the ranking, and two strategies including "DE/rand-to-Gbest/1" and "DE/rand/1" are adopted for the mutation operation. Then the proposed $(\mu + \lambda)$-MODE employs the binary crossover operation to generate the trial vectors via a crossover control parameter pool. In addition, $(\mu + \lambda)$-MODE integrates the fast nondominated sorting approach with the $(\mu + \lambda)$ selection strategy to choose the offspring in the evolutionary selection process. Eventually, experimental comparative results reveal that our proposed $(\mu + \lambda)$-MODE is superior to other EE methods in three real HSIs.

The rest of this article is organized as follows. In Section II, the related work about the LSMM, the traditional DE, multiobjective optimization is introduced. In Section III, we propose $(\mu + \lambda)$-MODE in detail. In Section IV, the experimental results are evaluated on three real HSIs. Eventually, Section V makes a conclusion and draws a future plan.

II. Related Work

A. LSMM

The LSMM [27] is commonly applied into the EE problem. Assume that the number of endmember is P for the HSI with N pixels and L bands. The pixel vectors of HSI are formulated as follows:

$$\mathbf{R} = \mathbf{E}\mathbf{A}^T + \varepsilon \tag{1}$$

where $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_P]$ represents the $L \times P$-dimensional endmember matrix. $\mathbf{R}$ stands for the $L \times N$-dimensional image. ε represents the noise and error matrix. $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_P]$ denotes the $N \times P$ dimensional abundance matrix.

The abundance has two constraints according to the practical meanings of abundance fractions. Two constraints are the abundance sum-to-one constraint (ASC), $\sum_{i=1}^{P} \mathbf{a}_i = \mathbf{1}$, and the abundance nonnegative constraint (ANC), $\mathbf{a}_i \geq \mathbf{0}, i = 1, 2, \ldots, P$. There are four kinds of abundance estimation strategies according to the constraints imposed on the abundance, namely, the fully constrained least squares (Fcls), the sum-to-one constrained least squares (Scls), the unconstrained least squares (Ucls), and the nonnegative constrained least squares (Ncls).

B. DE

DE is introduced by Price [21], Storn [21], and Storn and Price [23] and has turned into a significant algorithm among the evolutionary computation field [28]–[30]. The traditional DE contains four basic operations including the population initialization, mutation, crossover, and selection. As illustrated in Algorithm 1, the population is randomly obtained in the initialization process (lines 1 and 2). In the mutation operation (lines 5 and 6), the mutant vectors are obtained from the target vector of individuals via a variety of mutation strategies. After the mutation process, the binary crossover (lines 7–15) is commonly utilized in which some elements of the mutation vector can be updated by the individual vector to obtain the trial vector. Eventually, trial vectors or target vectors are chosen to remain offspring by the selection operation (lines 16–20). The traditional DE algorithm is illustrated as follows.

Algorithm 1 DE Algorithm

1: **Step 1: Initialization**
2: Randomly initialize the population and predefine the generation $G = 0$;
3: **while** the stopping condition is not satisfied **do**
4: **for** $i = 1$ to NP **do**
5: **Step 2: Mutation**
6: Obtain the mutant vector by the "DE/rand/1" strategy: $\vec{v}_{i,G} = x_{r_1,G} + F * (x_{r_2,G} - x_{r_3,G})$;
7: **Step 3: Crossover**
8: Reproduce the trial vector by the binary crossover:
9: **for** $j = 1$ to D **do**
10: **if** $(j_{rand} == j)\text{or}\left(rand_{i,j}[0, 1] \leq CR\right)$ **then**
11: $\vec{u}_{i,G}^{\,j} = \vec{v}_{i,G}^{\,j}$;
12: **else**
13: $\vec{u}_{i,G}^{\,j} = x_{i,G}^{j}$;
14: **end if**
15: **end for**
16: **Step 4: Selection**
17: Select the trial vector or the target vector by the fitness function value:
18: **if** $f(x_{i,G}) \geq f(\vec{u}_{i,G})$ **then**
19: $x_{i,G+1} = \vec{u}_{i,G}$;
20: **end if**
21: **end for**
22: $G = G + 1$;
23: **end while**

C. Multiobjective Optimization

Considering a minimization optimization problem, the MOP [31], [32] can be described as follows:

$$\begin{aligned} &\min F(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})] \\ &\text{s.t. } \mathbf{x} \in \Omega \end{aligned} \tag{2}$$

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where m is the objective number, Ω represents the decision space, and $\mathbf{x}$ denotes the decision variable. Due to the conflicts of objectives, the domination relationship is an effective method that can trade off objectives. When the vector u^* dominates the vector u, which is expressed as $u^* \prec u$, if and only if

$$\begin{cases} \forall i \in \{1, 2, \ldots, m\}, & f_i(u^*) \leq f_i(u) \\ \exists j \in \{1, 2, \ldots, m\}, & f_j(u^*) < f_j(u). \end{cases} \tag{3}$$

The u^* set where no other u in the decision space dominates u^* is called the Pareto set (PS). And the objective function set of the PS is defined as the Pareto front (PF).

Based on the domination relationship, NSGAII is a competitive MOEA [33]. In NSGAII, the fast nondominated sorting approach is the main operation that includes nondominated sorting and crowding distance sorting. As shown in Algorithm 2, in the process of nondominated sorting (lines 1–29), each individual is classified into the corresponding PF based on Pareto domination. After all PFs are assigned, the individuals of each PF are sorted by the crowding distance (lines 30–36). The fast nondominated sorting approach can be illustrated as follows.

Algorithm 2 Fast Nondominated Sorting Approach

1: **Step 1: non-dominated sorting**
2: $Front_1 = \emptyset$;
3: **for** every $x \in \mathbf{X}$ **do**
4: $D_x = \emptyset$; $n_x = 0$;
5: **for** every $x' \in \mathbf{X}$ **do**
6: **if** $x \prec x'$ **then**
7: $D_x \leftarrow D_x \cup x'$;
8: **else if** $x' \prec x$ **then**
9: $n_x = n_x + 1$;
10: **end if**
11: **end for**
12: **if** $n_x = 0$ **then**
13: $Front_1 \leftarrow Front_1 \cup x$;
14: **end if**
15: **end for**
16: $i = 1$;
17: **while** $Front_i \neq \emptyset$ **do**
18: $A = \emptyset$;
19: **for** every $x \in Front_i$ **do**
20: **for** every $x' \in D_x$ **do**
21: $n_{x'} = n_{x'} - 1$;
22: **if** $n_{x'} = 0$ **then**
23: $A \leftarrow A \cup x'$;
24: **end if**
25: **end for**
26: **end for**
27: $i = i + 1$;
28: $Front_i = A$;
29: **end while**
30: **Step 2: crowding distance sorting**
31: **for** every $Front_i$ **do**
32: **for** every $x \in Front_i$ **do**
33: Calculate the crowding distance of the individual x;
34: **end for**
35: Sort the individual of the PF $Front_i$ via the descending order of crowding distance;
36: **end for**

III. $(\mu + \lambda)$ Multiobjective DE Method Based on Ranking Multiple Mutations for Hyperspectral Endmember Extraction

A. Multiobjective Functions for EE

The EE is described as the MOP. The multiobjective functions are the volume inverse and the RMSE. They are listed as follows:

$$\begin{aligned} \min: F(\mathbf{X}) &= (f_1(\mathbf{X}), f_2(\mathbf{X})) \\ f_1(\mathbf{X}) &= \frac{1}{\text{volume}(\mathbf{Q}(\mathbf{X}))} = \frac{(P-1)!}{\left|\det\begin{bmatrix} 1 & 1 & \ldots & 1 \\ \mathbf{q}_1 & \mathbf{q}_2 & \ldots & \mathbf{q}_P \end{bmatrix}\right|} \\ f_2(\mathbf{X}) &= \text{RMSE}(\mathbf{r}, \hat{\mathbf{r}}) = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{1}{L}\|\mathbf{r}_i - \hat{\mathbf{r}}_i\|_2^2} \end{aligned} \tag{4}$$

where $\mathbf{Q}(\mathbf{X})$ denotes the endmember matrix, $\mathbf{X}$ represents the indices of endmembers in the HSI, P denotes the endmember sum, N represents the pixel total, L is the spectral dimensionality of the HSI, $\mathbf{r}$ and $\hat{\mathbf{r}}$ represent the pixel vector of the original image and the remixed image, respectively. The remixed pixel vector is obtained as follows:

$$\hat{\mathbf{r}}_i = \sum_{j=1}^{P} \hat{\alpha}_{i,j}\mathbf{e}_j \tag{5}$$

$$\mathbf{e} = \mathbf{r}(\mathbf{X}) \tag{6}$$

where abundances $\hat{\alpha}_{i,j}$ can be obtained by the Ncls [34], [35]. $\mathbf{e}$ is the endmember set.

B. Population Initialization

At first step, $(\mu + \lambda)$-MODE randomly initializes every individual in the population $\mathbf{X} = \{x_1, \ldots, x_i, \ldots, x_{\text{NP}}\}$, where x_i represents the ith individual and NP represents the population size. Every individual can be initialized as follows:

$$x_i = \text{randperm}(N, P) \tag{7}$$

where the function randperm(N, P) randomly generates a vector that contains P different integers in the range of $[1, N]$, N represents the pixel total in the HSI and P is the endmember total. Every individual $x_i = \{x_{i,1}, \ldots, x_{i,j}, \ldots, x_{i,P}\}$ denotes a candidate endmember set with P endmembers. The integer encoding of the individuals is shown in Fig. 1.

C. Ranking Multiple Mutations and Crossover Operator

After initialization, $(\mu + \lambda)$-MODE adopts ranking multiple mutations to create the mutant vectors. In the ranking multiple mutations, two differential mutations are described as follows.

"DE/rand/1":

$$\vec{v}_{i_1,G} = x_{r_1,G} + F * (x_{r_2,G} - x_{r_3,G}). \tag{8}$$

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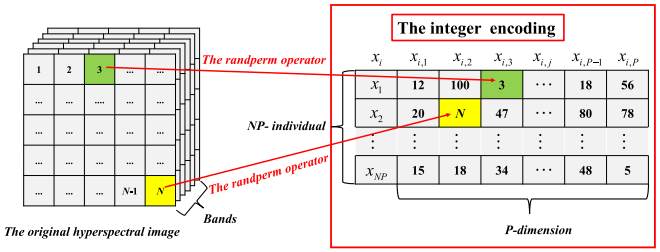


Fig. 1. Integer encoding of the individuals.

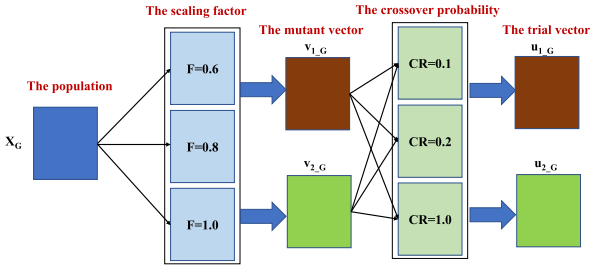


Fig. 2. Illustration of generating the trial vector by the parameter candidate pool.

"DE/rand-to-Gbest/1":

$$\vec{v}_{i_2,G} = x_{r_1,G} + \text{rand} * (x_{\text{Gbest},G} - x_{r_1,G}) + F * (x_{r_2,G} - x_{r_3,G}) \tag{9}$$

where F represents the mutation scaling factor, G denotes the current generation, rand represents a random value within [0, 1], $x_{\text{Gbest},G}$ is stochastically selected from the obtained PS (Gbest), $\vec{v}_{i_1,G}$ and $\vec{v}_{i_2,G}$ are mutation vectors generated via the ranking multiple mutations. Inspired by Gong and Cai [36] and Gong *et al.* [37], three different indices r_1, r_2, r_3 can be randomly assigned by the ranking-based vector selection. The ranking degree of the ith individual is calculated as follows:

$$\text{Rank}_i = \text{NP} - i. \tag{10}$$

The corresponding selection probability is obtained as follows:

$$p_i = \left(\frac{\text{Rank}_i}{\text{NP}}\right)^2. \tag{11}$$

The individual vectors of the higher selection probability have more chances to be selected. The ranking-based vector selection can be given in Algorithm 3.

Algorithm 3 Ranking-Based Vector Selection

```
Randomly generate r1 ∈ {1, 2, ..., NP};
while r1 == i or rand > p_{r1} do
    Randomly reelect r1 ∈ {1, 2, ..., NP};
end while
Randomly generate r2 ∈ {1, 2, ..., NP};
while r2 == i or r2 == r1 or rand > p_{r2} do
    Randomly reelect r2 ∈ {1, 2, ..., NP};
end while
Randomly generate r3 ∈ {1, 2, ..., NP};
while r3 == r1 or r3 == r2 or r3 == i do
    Randomly reelect r3 ∈ {1, 2, ..., NP};
end while
```

The $(\mu+\lambda)$-MODE applies the traditional binomial crossover operator to obtain the trial vectors. The crossover operators are shown as follows:

$$\vec{u}_{i_1,G}^{\,j} = \begin{cases} \vec{v}_{i_1,G}^{\,j} & if(j == j_{\text{rand}}) || (\text{rand}_j[0,1] \le CR) \\ x_{i,G}^{j} & \text{otherwise} \end{cases}$$
$$\vec{u}_{i_2,G}^{\,j} = \begin{cases} \vec{v}_{i_2,G}^{\,j} & if(j == j_{\text{rand}}) || (\text{rand}_j[0,1] \le CR) \\ x_{i,G}^{j} & \text{otherwise} \end{cases} \tag{12}$$

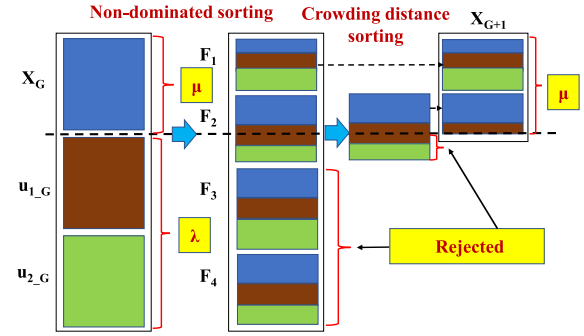


Fig. 3. Fast nondominated sorting approach and the $(\mu+\lambda)$ selection strategy.

Algorithm 4 Reasonable Reset Approach

```
for k = 1 : 2 do
    for j = 1 : P do
        Step 1: Out-of-bounds processing
        if u^j_{i_k,G} < X_min || u^j_{i_k,G} > X_max then
            u^j_{i_k,G} = X_min + rand * (X_max − X_min);
        end if
        Step 2: Rounding
        u^j_{i_k,G} = round(u^j_{i_k,G});
    end for
    Step 3: De-duplication
    temp = unique(u^j_{i_k,G});
    if length(temp) < P then
        pos = [X_min : X_max];
        pos(temp) = [ ];
        num = randperm(length(pos), P − length(temp));
        temp = [temp, pos(num)];
    end if
    u^j_{i_k,G} = temp;
end for
```

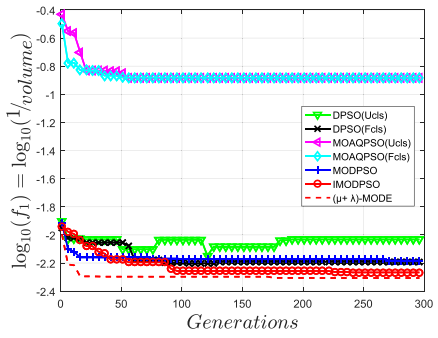
where j_{rand} is a random integer within $[1, P]$ and CR is the crossover probability. Motivated by Wang *et al.* [38], [39], the scaling factor pool ($F_{\text{pool}} = [0.6, 0.8, 1.0]$) and the crossover probability pool ($CR_{\text{pool}} = [0.1, 0.2, 1.0]$) can be constructed to randomly choose F and CR. Illustration of generating trial vectors by the parameter candidate pool is shown in Fig. 2.

In order to ensure the rationality of the trial vector, the reasonable reset approach is adopted. The reasonable reset approach consists of three steps as shown in Algorithm 4.

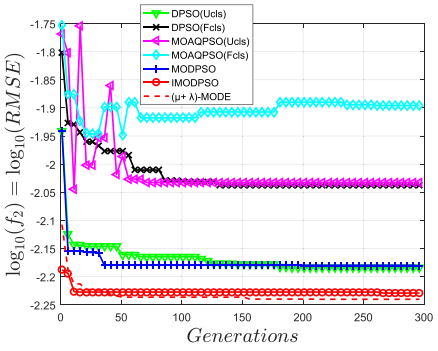
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Fig. 4. False-color image of Samson.



(a)



(b)

Fig. 5. Convergence curves for the Samson data set. (a) Logarithm of volume inverse. (b) Logarithm of RMSE.

1) Out-of-bounds processing (lines 3–6). If the trial vector value is out-of-bounds, the trial vector value will be randomly initialized.
2) Rounding (lines 7 and 8). After out-of-bounds processing, rounding is applied.
3) Deduplication (lines 10–18). If the trial vector has repeated values, deduplication will be utilized and the repeated values will be replaced by different values.

D. Fast Nondominated Sorting Approach and the $(\mu + \lambda)$ Selection Strategy

After the crossover process, the combination of the current population and the trial vectors are sorted by the fast nondominated sorting approach [33], which contains nondominated sorting and crowding distance sorting. Each PF is obtained by the non-dominated sorting. Afterward individuals of every PF are arranged via the crowding distance. Next, the next generation individuals are survived via the $(\mu + \lambda)$ selection strategy [40], [41]. In the $(\mu + \lambda)$ selection process, the $(\mu + \lambda)$ combination set selects μ individuals as the next parent individuals and the remaining λ individuals are rejected into the next generation. Compared with trial vectors and individuals, Fig. 3 illustrates the selection process.

TABLE I
OBJECTIVE RESULTS FOR THE SAMSON DATA SET

Algorithms	$f_1 = 1/volume$	$f_2 = RMSE$	Time(sec)
N-FINDR	4.23E-03	1.01E-02	1.09E+00
VCA	4.35E-03	2.35E-02	5.31E-01
MOAQPSO(Ucls)	1.31E-01	9.27E-03	9.46E+02
MOAQPSO(Fcls)	1.31E-01	1.27E-02	2.37E+03
DPSO(Ucls)	9.29E-03	6.55E-03	3.47E+02
DPSO(Fcls)	6.44E-03	9.18E-03	3.85E+04
MODPSO	6.57E-03 6.58E-03 6.85E-03 6.87E-03 7.24E-03 7.40E-03 7.59E-03 7.78E-03 7.80E-03 8.34E-03 8.75E-03 8.79E-03 8.98E-03 9.02E-03 9.49E-03	7.73E-03 7.41E-03 7.08E-03 7.07E-03 7.06E-03 7.04E-03 7.02E-03 6.88E-03 6.87E-03 6.72E-03 6.63E-03 6.62E-03 6.61E-03 6.61E-03 6.59E-03	3.63E+02
IMODPSO	5.41E-03 5.41E-03 6.14E-03 6.14E-03 6.16E-03 6.18E-03 6.65E-03 6.66E-03 7.66E-03 7.82E-03	6.89E-03 6.84E-03 6.43E-03 6.40E-03 6.39E-03 6.37E-03 5.97E-03 5.93E-03 5.90E-03 5.90E-03	1.48E+04
$(\mu+\lambda)$-MODE	4.93E-03 4.94E-03 4.95E-03 5.18E-03 5.35E-03 5.37E-03 5.76E-03 5.96E-03 6.06E-03 6.18E-03 6.36E-03 6.41E-03 6.58E-03 6.77E-03 6.77E-03 7.10E-03 7.14E-03 7.32E-03 7.35E-03 7.36E-03 8.10E-03 8.13E-03	7.23E-03 7.20E-03 7.16E-03 7.01E-03 6.81E-03 6.73E-03 6.64E-03 6.57E-03 6.33E-03 6.19E-03 6.06E-03 5.97E-03 5.90E-03 5.88E-03 5.87E-03 5.85E-03 5.82E-03 5.80E-03 5.78E-03 5.76E-03 5.75E-03 5.75E-03	2.88E+04

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Overall, the framework of $(\mu+\lambda)$-MODE can be given in Algorithm 5. According to [33], the overall computational complexity of NSGAII is $O(G_{\max}\cdot m\cdot(\mathrm{NP})^2)$, where $G_{\max}$ is the maximum generation iteration, m is the objective number and NP is the population size. The framework of $(\mu+\lambda)$-MODE is similar to NSGAII. Different from NSGAII, $(\mu+\lambda)$-MODE just extends ranking multiple mutations, crossover operator and the $(\mu+\lambda)$ selection strategy. So it does not increase the computation complexity. The computational complexity of $(\mu+\lambda)$-MODE is $O(G_{\max}\cdot m\cdot(\mathrm{NP})^2)$.

Algorithm 5 $(\mu+\lambda)$-Mode

Input: The population number NP, the maximum generation max_gen, the endmember number P, the crossover probability pool CR_{pool} and the scaling factor pool F_{pool}.

Output: The obtained Pareto set $(Gbest)$.

$G=0$;
Utilize a random initialization of the population $\mathbf{X}_G=\{x_{1,G},x_{2,G},\ldots,x_{NP,G}\}$;
Evaluate the objective functions of the population;
Utilize the fast non-dominated sorting approach to sort $\mathbf{X}_G$;
Select the obtained Pareto set $(Gbest)$ from the population;
while $G<max_gen$ **do**
  $\mathbf{X}_{G+1}=\emptyset$;
  for $i=1:NP$ **do**
    Stochastically choose a control parameter setting on the parameter candidate pool;
    Utilize the ranking-based vector selection to determine the indexes r_1,r_2,r_3;
    Generate two mutant vectors $\vec{v}_{i_1,G}$ and $\vec{v}_{i_2,G}$ by the multiple mutation strategy;
    Generate two trial vectors $\vec{u}_{i_1,G}$ and $\vec{u}_{i_2,G}$ using crossover operator;
    Perform the reasonable reset approach to ensure the rationality of trial vectors;
    Evaluate the objective functions of trial vectors;
    $\mathbf{X}_{G+1}=\mathbf{X}_G\cup\vec{u}_{i_1,G}\cup\vec{u}_{i_2,G}$;
  end for
  Utilize the fast non-dominated sorting approach to sort $\mathbf{X}_{G+1}$;
  Select the obtained Pareto set $(Gbest)$ from the population;
  if $|\mathbf{X}_{G+1}|>NP$ **then**
    Utilize the $(\mu+\lambda)$ selection strategy, delete the last $(|\mathbf{X}_{G+1}|-NP)$ individuals in $\mathbf{X}_{G+1}$;
  end if
  $G{+}{+}$;
end while

IV. Performance Evaluation

The proposed $(\mu+\lambda)$-MODE method is evaluated on three real hyperspectral images. Compared with $(\mu+\lambda)$-MODE, VCA [13], and N-FINDR [11] are conventional EE algorithms, and MOAQPSO [20], DPSO [17], MODPSO [25], and IMODPSO [26] are EA-based EE algorithms. Especially, MODPSO and IMODPSO are MOEAs for EE. In the experiments, MOAQPSO with Ucls and Fcls are denoted by MOAQPSO (Ucls) and MOAQPSO (Fcls), respectively, so does DPSO with Ucls and Fcls. The population size is 20, and the maximum generation is 300. In the DPSO and MODPSO, the random selection probability is predefined as 0.2. The parameter settings of $(\mu+\lambda)$-MODE are $F_{\mathrm{pool}}=[0.6,0.8,1.0]$ and $CR_{\mathrm{pool}}=[0.1,0.2,1.0]$. The evaluation criterions of these methods are the RMSE and the volume inverse.

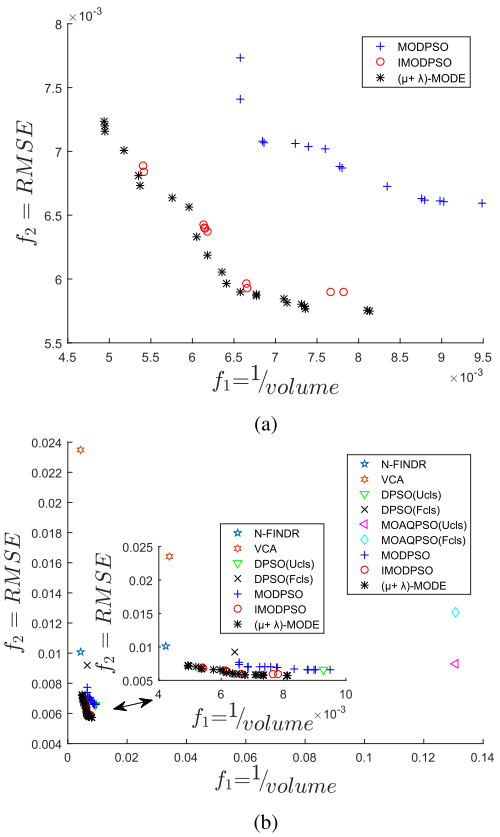


Fig. 6. Results of the Samson image. (a) PF comparison results by three multiobjective methods. (b) Results obtained by all algorithms.

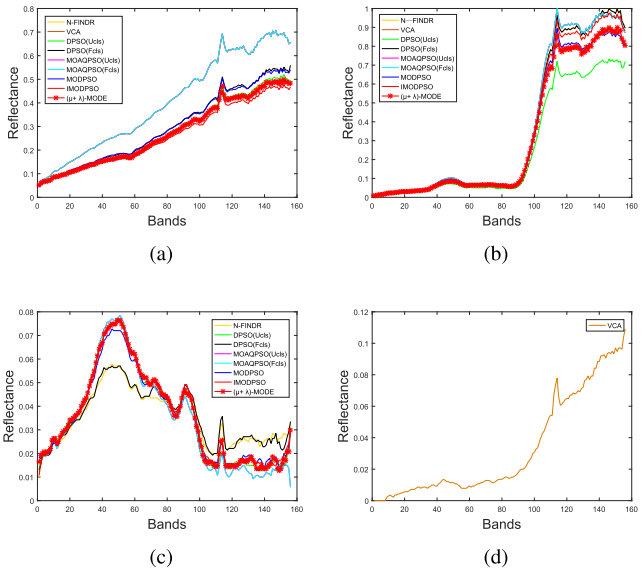


Fig. 7. Endmember spectra extraction results via all algorithms for Samson data set. (a) Soil. (b) Tree. (c) Water. (d) Unmatched spectra.

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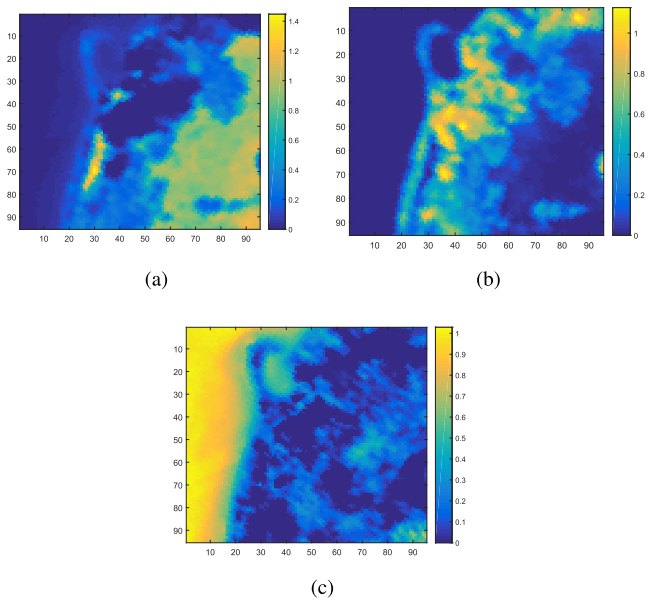


Fig. 8. Abundance of $(\mu + \lambda)$-MODE generated by the Ncls for Samson data set. (a) Soil. (b) Tree. (c) Water.

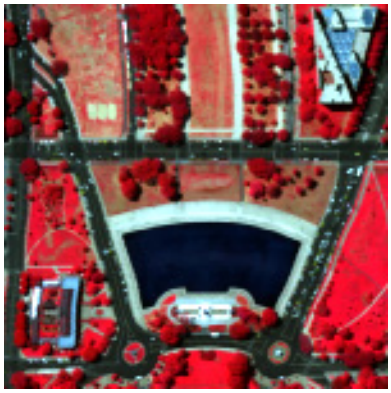


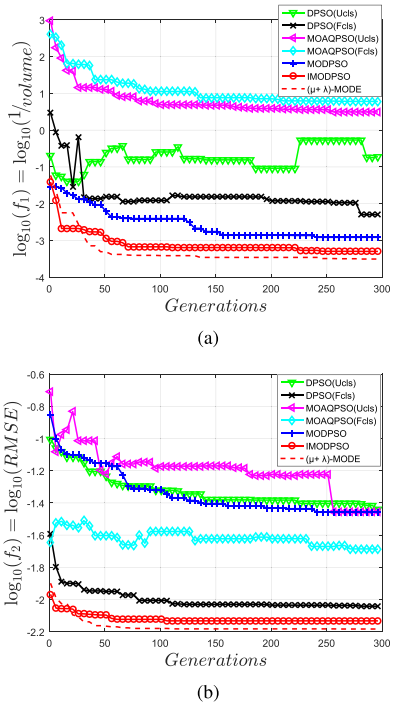
Fig. 9. False-color image of Washington DC.

A. Samson Data Set Experimental Results

In the Samson data set, there are 952 × 952 pixels, each of which contains 156 bands. The subset of the Samson data set used here has 95 × 95 pixels. The false-color image is listed in Fig. 4, with band 64 being red, band 52 being green and band 36 being blue. The endmembers existed in this image are soil, tree, and water [42]–[44].

For the Samson data set, all algorithms are evaluated via minimizing the volume inverse and the RMSE. In Table I, the object results illustrate that in aspect of the volume inverse, $(\mu + \lambda)$-MODE outperforms other evolutionary-based algorithms apart from the traditional EE algorithms. Also, in the aspect of the RMSE, $(\mu + \lambda)$-MODE performs the best among all comparative algorithms. Even if $(\mu + \lambda)$-MODE costs the more computation time, $(\mu + \lambda)$-MODE is still valid for the accuracy of the volume inverse and the RMSE.

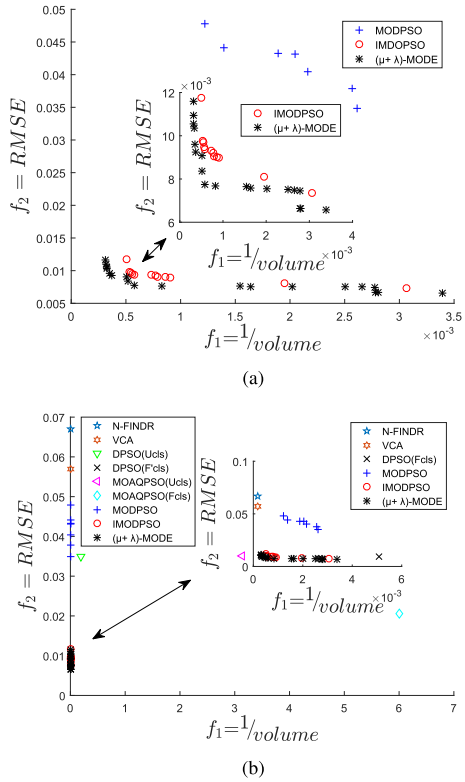
Convergence is the crucial metric for evolutionary-based algorithms. In Fig. 5, x-axis indicates the generation number, and y-axis of Fig. 5(a) or (b) stands for the logarithm of



(a)

(b)

Fig. 10. Convergence curves for the Washington DC data set. (a) Logarithm of volume inverse. (b) Logarithm of RMSE.



(a)

(b)

Fig. 11. Results of the Washington DC image. (a) PF comparison results by three multiobjective methods. (b) Results obtained by all algorithms.

the volume inverse or the RMSE, respectively. In terms of the volume inverse, $(\mu + \lambda)$-MODE converges faster than DPSO (Ucls and Fcls), MOAQPSO (Ucls and Fcls), MODPSO

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TABLE II
MINIMUM SAD RESULT FOR THE SAMSON DATA SET

	N-FINDR	VCA	MOAQPSO(Ucls)	MOAQPSO(Fcls)	DPSO(Ucls)	DPSO(Fcls)	MODPSO	IMODPSO	$(\mu+\lambda)$-MODE
1-soil	0.040435158	0.040435158	0.040435158	0.040435158	0.01595176	0.02064841	0.01309301	0.0060274	0.008762157
2-tree	0.040685317	0.021904421	0.021904421	0.021904421	0.03533967	0.04068532	0.01733827	0.0363564	0.033284874
3-water	0.160987248	*	0.12958521	0.12958521	0.06904661	0.15949347	0.04326994	0.0488752	0.06314166

TABLE III
OBJECTIVE RESULTS FOR THE WASHINGTON DC DATA SET

Algorithms	$f_1 = 1/volume$	$f_2 = RMSE$	Time(sec)
N-FINDR	1.78E-04	6.70E-02	2.42E+00
VCA	1.87E-04	5.70E-02	1.53E+00
MOAQPSO(Ucls)	3.13E+00	3.51E-02	3.52E+03
MOAQPSO(Fcls)	6.01E+00	2.05E-02	1.69E+04
DPSO(Ucls)	1.89E-01	3.49E-02	7.51E+02
DPSO(Fcls)	5.07E-03	9.08E-03	9.06E+04
MODPSO	1.22E-03	4.78E-02	7.14E+02
	1.39E-03	4.41E-02	
	1.89E-03	4.32E-02	
	2.04E-03	4.32E-02	
	2.16E-03	4.04E-02	
	2.57E-03	3.78E-02	
	2.61E-03	3.48E-02	
IMODPSO	5.06E-04	1.18E-02	4.47E+04
	5.32E-04	9.76E-03	
	5.48E-04	9.70E-03	
	5.59E-04	9.46E-03	
	5.74E-04	9.39E-03	
	7.33E-04	9.32E-03	
	7.73E-04	9.22E-03	
	7.87E-04	9.03E-03	
	8.60E-04	9.03E-03	
	9.07E-04	8.98E-03	
	1.95E-03	8.12E-03	
	3.06E-03	7.37E-03	
$(\mu+\lambda)$-MODE	3.14E-04	1.16E-02	8.66E+04
	3.20E-04	1.09E-02	
	3.22E-04	1.05E-02	
	3.28E-04	1.04E-02	
	3.48E-04	9.60E-03	
	3.72E-04	9.25E-03	
	5.09E-04	9.07E-03	
	5.20E-04	8.37E-03	
	5.76E-04	7.74E-03	
	8.26E-04	7.67E-03	
	1.54E-03	7.66E-03	
	1.64E-03	7.57E-03	
	2.02E-03	7.56E-03	
	2.51E-03	7.52E-03	
	2.66E-03	7.50E-03	
	2.78E-03	7.44E-03	
	2.78E-03	6.64E-03	
	2.81E-03	6.63E-03	
	3.39E-03	6.56E-03	

and IMODPSO. In terms of the RMSE, $(\mu + \lambda)$-MODE also achieves the highest convergence speed among all comparative evolutionary-based algorithms. As the number of generation reaches 50, $(\mu + \lambda)$-MODE surpasses IMODPSO on the convergence speed.

Fig. 6 displays the PF results. From Fig. 6(a), the PF comparison of three MOEAs reveals that the whole solutions of MODPSO and IMODPSO are dominated by the most solutions of $(\mu + \lambda)$-MODE. The results of all algorithms

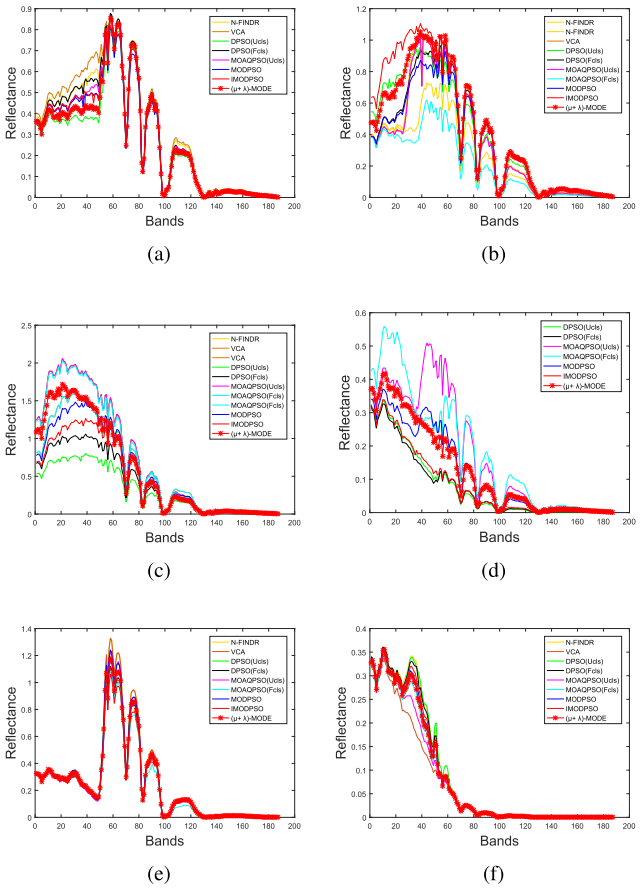


Fig. 12. Endmember spectra extraction results via all algorithms for Washington DC data set. (a) Grass. (b) Path. (c) Roof. (d) Street. (e) Tree. (f) Water.

are displayed in Fig. 6(b). Compared with $(\mu + \lambda)$-MODE, dominated solutions are obtained by DPSO (Ucls and Fcls) and MOAQPSO (Ucls and Fcls), while VCA and N-FINDR obtain the nondominated solutions. Overall, $(\mu + \lambda)$-MODE achieves the high-quality PF for the Samson image.

The extracted endmember spectra are shown in Fig. 7 and the minimum value of the well-known spectral angle distance (SAD) values of each algorithm is displayed in Table II. Though VCA fails to extract the water endmember, most algorithms can successfully extract the soil endmember, the tree endmember, and the water endmember. Except for IMODPSO, $(\mu + \lambda)$-MODE is also competitive and obtains the relatively small SAD value for the soil endmember.

The abundance maps obtained by $(\mu+\lambda)$-MODE are shown in Fig. 8.

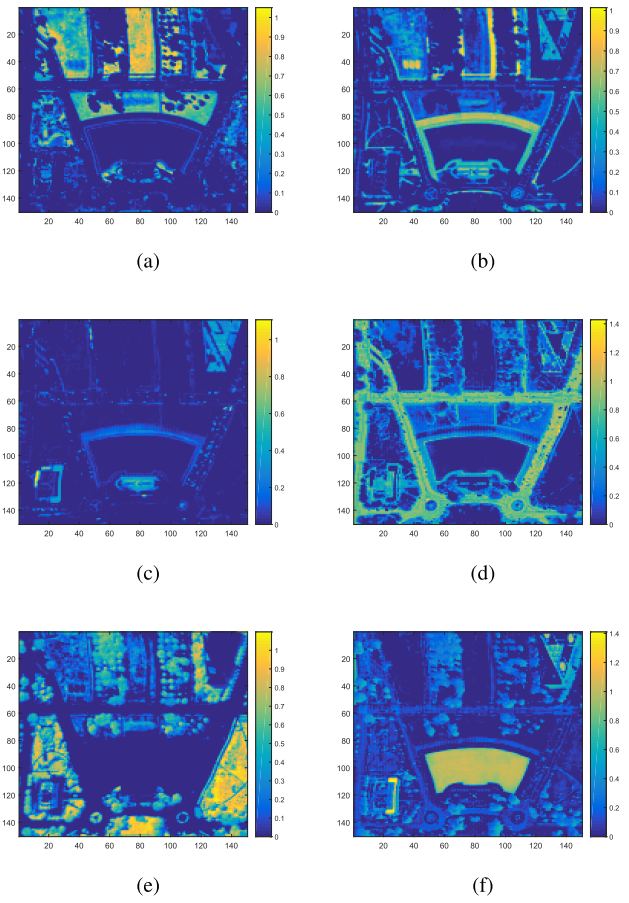
B. Washington DC Data Set Experimental Results

The Washington DC data set is collected by HYDICE sensor. The subscene is extracted from the original Washington

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TABLE IV
MINIMUM SAD RESULT FOR THE WASHINGTON DC DATA SET

	N-FINDR	VCA	MOAQPSO(Ucls)	MOAQPSO(Fcls)	DPSO(Ucls)	DPSO(Fcls)	MODPSO	IMODPSO	$(\mu+\lambda)$-MODE
1-grass	0.106982378	0.148361205	0.067146636	*	0.046221044	0.087320514	0.07551154	0.060620773	0.026640585
2-path	0.164401259	0.181233806	0.181233806	0.179019099	0.072372169	0.117888948	0.106600111	0.087349729	0.03450803
3-roof	0.105701401	0.105701401	0.105701401	0.100339378	0.192845627	0.184129454	0.20597235	0.220985982	0.073360727
4-street	*	*	0.295060922	0.179061613	0.201639687	0.206471607	0.1393668	0.167192292	0.017851985
5-tree	0.174246132	0.174246132	0.170368292	0.119109493	0.144681654	0.138341643	0.152910756	0.125208917	0.136871819
6-water	0.070836817	0.134163644	0.071009262	0.013637892	0.076559115	0.05229557	0.019870986	0.032620675	0.008767442



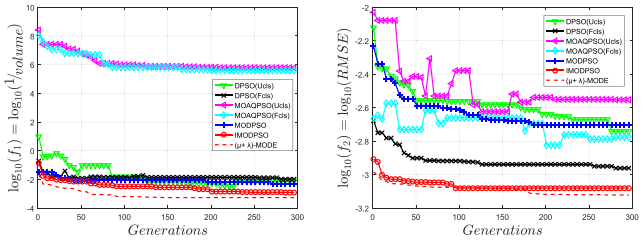
(a) (b) (c) (d) (e) (f)

Fig. 13. Abundance of $(\mu+\lambda)$-MODE generated by the Ncls for Washington DC data set. (a) Grass. (b) Path. (c) Roof. (d) Street. (e) Tree. (f) Water.



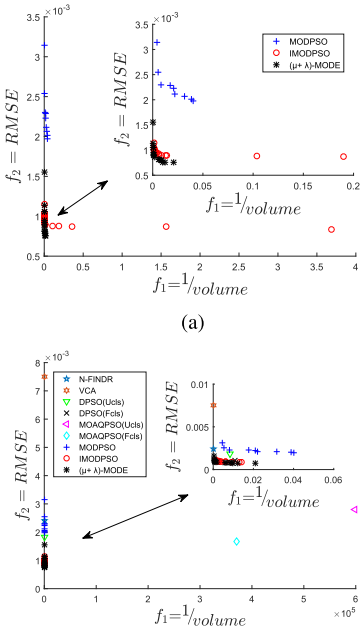
Fig. 14. False-color image of Urban.

DC image and has 150 × 150 pixels covering 187 bands. The false-color image of Washington DC is displayed in Fig. 9. Six endmembers contain grass, path, roof, street, tree, and water in the subset image [45].



(a) (b)

Fig. 15. Convergence curves for the Urban data set. (a) Logarithm of volume inverse. (b) Logarithm of RMSE.



(a) (b)

Fig. 16. Results of the Urban image. (a) PF comparison results by three multiobjective methods. (b) Results obtained by all algorithms.

On the Washington DC data set, Table III illustrates the objective results of all algorithms. On the minimization of the volume inverse, $(\mu+\lambda)$-MODE outperforms other EA-based EE algorithms except for the traditional algorithms N-FINDR and VCA. According to the RMSE results, $(\mu+\lambda)$-MODE obtains the minimum value. The drawback of $(\mu+\lambda)$-MODE lies in the high costs of computation time.

From the convergence curves in Fig. 10, we can observe that with more than 25 generations, $(\mu+\lambda)$-MODE surpasses IMODPSO on the volume inverse and the RMSE. After that, $(\mu+\lambda)$-MODE has the best convergence speed among all comparative EA-based EE algorithms.

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TABLE VI
MINIMUM SAD RESULT FOR THE URBAN DATA SET

	N-FINDR	VCA	MOAQPSO(Ucls)	MOAQPSO(Fcls)	DPSO(Ucls)	DPSO(Fcls)	MODPSO	IMODPSO	$(\mu+\lambda)$-MODE
1-road#1	*	*	*	*	0.034965237	0.076753076	0.096539258	0.074683106	0.075950332
2-roof#1	*	*	*	*	*	0.153696843	0.027115762	0.021096915	0.033512329
3-grass	0.089777582	*	0.075404558	0.078793934	0.144169056	0.062745647	0.07454709	0.065393005	0.066809685
4-tree	0.110090352	0.049045106	0.07730557	0.116153936	0.140359767	0.106004184	0.090776893	0.05898219	0.092022362
5-road#2	*	*	0.122795029	0.165055767	*	0.081877395	0.081249295	0.110532491	0.124274339
6-roof#2	0.267765262	0.267765262	*	0.178375666	0.029130453	*	0.247993726	*	0.052945918

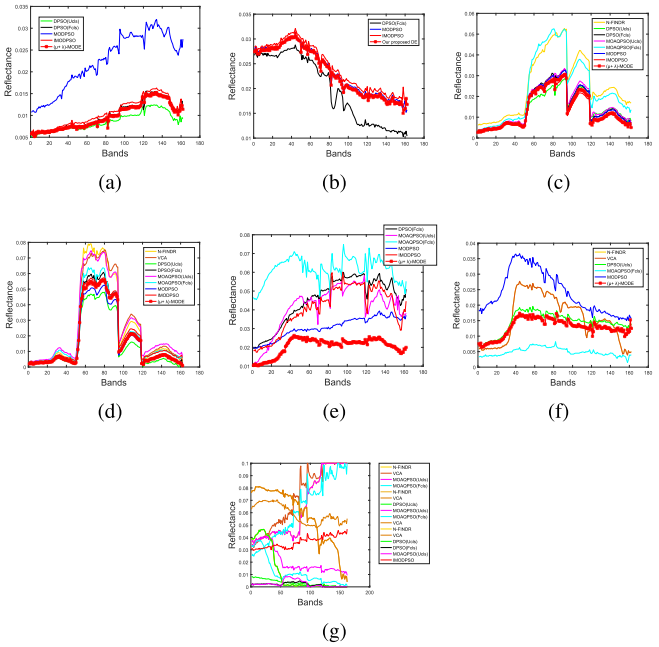


Fig. 17. Endmember spectra extraction results via the all algorithms for Urban data set. (a) Road#1. (b) Roof#1. (c) Grass. (d) Tree. (e) Road#2. (f) Roof#2. (g) Unmatched spectra.

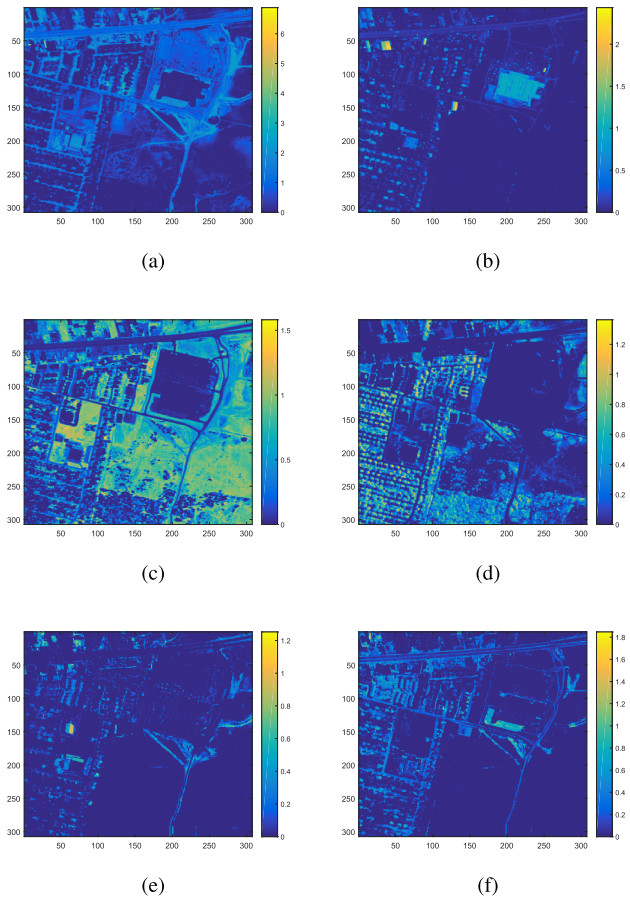


Fig. 18. Abundance of $(\mu + \lambda)$-MODE generated by the Ncls for Urban data set. (a) Road#1. (b) Roof#1. (c) Grass. (d) Tree. (e) Road#2. (f) Roof#2.

The PF results of all algorithms are displayed in Fig. 11. From Fig. 11(a), the PF results of three MOEAs are compared and we can observe that $(\mu + \lambda)$-MODE dominates MODPSO and IMODPSO. Though N-FINDR and VCA are nondominated to $(\mu + \lambda)$-MODE from Fig. 11(b), some of volume inverse values of $(\mu + \lambda)$-MODE are very close to that of VCA and N-FINDR, and the RMSE values of $(\mu + \lambda)$-MODE are much smaller than that of VCA and N-FINDR. In fact, $(\mu + \lambda)$-MODE obtains the high-quality PF of better diversity and dominates all comparative EA methods on the Washington DC data set.

Six endmember spectra are shown in Fig. 12. From Fig. 12, $(\mu + \lambda)$-MODE can successfully extract six different endmembers. The minimum SAD values of each algorithm are displayed in Table IV. In terms of SAD, $(\mu + \lambda)$-MODE can obtain the smallest value for the grass, path, roof, street and water endmember. In Fig. 13, the abundance map of $(\mu + \lambda)$-MODE is obtained by Ncls. From Fig. 13(a) to (f), the maps display the abundances of grass, path, roof, street, tree, and water.

C. Urban Data Set Experimental Results

The Urban image has 307 × 307 pixels with 162 bands. The false-color image is illustrated in Fig. 14, band 52 is set to green, band 36 is set to blue, and band 64 is set to red. Six endmembers contain road#1, roof#1, grass, tree, road#2 and roof#2 in HSI [46].

On the Urban data set, the comparison results of all methods are illustrated in Table V. We can observe that VCA and N-FINDR can obtain good results on the volume inverse, but are inferior to $(\mu + \lambda)$-MODE on the RMSE. Compared to other EA-based EE algorithms, $(\mu + \lambda)$-MODE obtains the best results on the RMSE and the volume inverse. In spite of expensive computation time, $(\mu + \lambda)$-MODE can improve the result of two objective function values.

The convergence curves are shown in Fig. 15. Compared with other EA-based EE algorithms, the convergence speed of $(\mu + \lambda)$-MODE is the fastest on two objectives.

The PF front results are illustrated in Fig. 16. We can observe in Fig. 16(a) that the most results of $(\mu + \lambda)$-MODE dominates the results of MODPSO and IMODPSO. Though the results of $(\mu + \lambda)$-MODE is nondominated to those of

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TABLE V
OBJECTIVE RESULTS FOR THE URBAN DATA SET

Algorithms	$f_1 = 1/volume$	$f_2 = RMSE$	Time(Sec)
N-FINDR	3.95E-05	2.41E-03	7.09E+00
VCA	4.28E-05	7.50E-03	3.48E+00
MOAQPSO(Ucls)	5.99E+05	2.80E-03	1.22E+04
MOAQPSO(Fcls)	3.70E+05	1.67E-03	6.15E+04
DPSO(Ucls)	8.40E-03	1.82E-03	3.53E+03
DPSO(Fcls)	1.07E-02	1.09E-03	4.53E+05
MODPSO	4.48E-03	3.14E-03	2.28E+03
	5.36E-03	2.54E-03	
	8.11E-03	2.30E-03	
	1.76E-02	2.29E-03	
	2.10E-02	2.23E-03	
	2.21E-02	2.11E-03	
	3.17E-02	2.06E-03	
	3.87E-02	2.01E-03	
	4.03E-02	1.97E-03	
IMODPSO	1.23E-03	1.15E-03	1.65E+05
	1.33E-03	1.04E-03	
	1.33E-03	1.04E-03	
	1.45E-03	1.01E-03	
	1.46E-03	1.00E-03	
	1.61E-03	9.90E-04	
	1.61E-03	9.88E-04	
	1.93E-03	9.85E-04	
	2.59E-03	9.72E-04	
	2.60E-03	9.69E-04	
	2.62E-03	9.68E-04	
	2.85E-03	9.55E-04	
	3.60E-03	9.49E-04	
	3.85E-03	9.35E-04	
	6.09E-03	9.33E-04	
	6.13E-03	9.10E-04	
	9.14E-03	8.99E-04	
	1.29E-02	8.98E-04	
	1.41E-02	8.89E-04	
	1.04E-01	8.79E-04	
	1.90E-01	8.76E-04	
	3.54E-01	8.72E-04	
	1.56E+00	8.69E-04	
	3.69E+00	8.31E-04	
$(\mu+\lambda)$-MODE	5.26E-04	1.55E-03	3.32E+05
	5.39E-04	1.13E-03	
	5.66E-04	1.06E-03	
	6.08E-04	1.05E-03	
	6.48E-04	9.52E-04	
	8.88E-04	9.19E-04	
	9.19E-04	8.91E-04	
	1.88E-03	8.61E-04	
	5.91E-03	8.15E-04	
	6.33E-03	8.11E-04	
	8.45E-03	8.04E-04	
	9.09E-03	7.88E-04	
	9.76E-03	7.61E-04	
	1.21E-02	7.58E-04	
	2.08E-02	7.56E-04	

N-FINDR and VCA in Fig. 16(b), $(\mu + \lambda)$-MODE can dominate the rest EA-based EE algorithms including MOAQPSO (Ucls and Fcls) and DPSO (Ucls and Fcls).

Fig. 17 shows the six endmember spectra and unmatched spectra. In the Urban data set, $(\mu + \lambda)$-MODE and MODPSO can extract six endmember spectra completely. The minimum SAD values of each algorithm are displayed in Table VI. On the endmembers of the road#1, the grass and the roof#2, $(\mu + \lambda)$-MODE can obtain the smaller SAD value than MODPSO. As shown in Fig. 18, the abundance maps of $(\mu + \lambda)$-MODE is generated by Ncls. From the abundance maps, we can observe road#1, roof#1, grass, tree, road#2, and roof#2.

V. CONCLUSION

The EE problem is an important task that can be considered as MOP. From our work, a $(\mu + \lambda)$ multiobjective DE method based on ranking multiple mutations [$(\mu+\lambda)$-MODE] is proposed for hyperspectral EE to optimize two objectives, namely, the RMSE and the volume inverse simultaneously. In the proposed $(\mu + \lambda)$-MODE algorithm, we adopt ranking multiple mutations with a scaling factor pool to generate mutant vectors. In the ranking multiple mutations process, parent vectors with higher ranking have more chances to join the mutation operations. Then $(\mu + \lambda)$-MODE employs the binary crossover operator with a crossover control parameter pool to obtain the trial vectors. Furthermore, so as to select the high-quality offspring population, we incorporate the fast nondominated sorting approach and the $(\mu + \lambda)$ selection strategy into $(\mu + \lambda)$-MODE. Experimental results on three HSIs verify the effectiveness of our proposed $(\mu + \lambda)$-MODE algorithm and show that $(\mu + \lambda)$-MODE obtains better EE results.

In future work, we will add new objectives into the EE problem which can be transformed into the many-objective optimization problem. Next, we will focus on the many-objective EA for EE problem. We will also aim at designing a better many-objective EA to reduce computation time.

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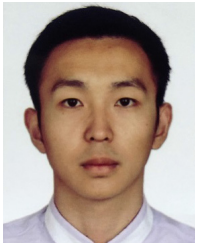
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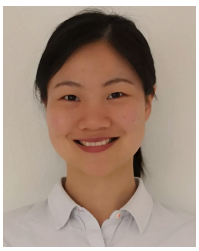
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